



RSANet: Relative-sequence quality assessment network for gait recognition in the wild

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ABSTRACT

Gait recognition in the wild has received increasing attention since the gait pattern is hard to disguise and can be captured in a long distance. However, due to occlusions and segmentation errors, low-quality silhouettes are common and inevitable. To mitigate this low-quality problem, some prior arts propose absolute-single quality assessment models. Although these methods obtain a good performance, they only focus on the silhouette quality of a single frame, lacking consideration of the variation state of the entire sequence. In this paper, we propose a Relative-Sequence Quality Assessment Network, named RSANet. It uses the Average Feature Similarity Module (AFSM) to evaluate silhouette quality by calculating the similarity between one silhouette and all other silhouettes in the same silhouette sequence. The silhouette quality is based on the sequence, reflecting a relative quality. Furthermore, RSANet uses Multi-Temporal-Receptive-Field Residual Blocks (MTB) to extend temporal receptive fields without parameter increases. It achieves a Rank-1 accuracy of 75.2% on Gait3D, 81.8% on GREW, and 77.6% on BUAA-Duke-Gait datasets respectively. The code is available at <https://github.com/PGZ-Sleepy/RSANet>.

1. Introduction

Gait recognition is a biometric technique that recognizes individuals from their body shapes and walk postures. Since gait is considered a silhouette pattern that can be captured at a distance without cooperation, it is promising to be used in surveillance scenarios, including criminal investigation, identity verification, and sports science.

Recently, appearance-based methods [1,2] using silhouette sequence as input have achieved successful performance on in-the-lab datasets, *i.e.*, 92.3% Rank-1 accuracy [1] on CASIA-B [3] and 90.7% Rank-1 accuracy [2] on OU-MVLP [4]. However, the performance of these appearance-based methods drops dramatically on in-the-wild datasets, *i.e.*, Gait3D [5] and GREW [6]. The primary reason for this performance degradation is straightforward: the scenario for capturing existing in-the-lab datasets differs greatly from in-the-wild datasets. Specifically, pedestrians on in-the-wild datasets may walk at various velocities and follow non-linear paths. In addition, pedestrians can be occluded by other pedestrians or objects. Such complex scenarios are not included in in-the-lab datasets. Therefore, models designed according to in-the-lab datasets are not reliable in wild scenarios, resulting in performance degradation.

To tackle the problem of gait recognition in wild scenarios, first of all, silhouettes on in-the-wild datasets [5,6] are analyzed. We observe that there exists many partial silhouettes missing. As illustrated in Fig. 1(a), the human trunk and limbs exhibit absence in the last several frames, due to incomplete segmentation caused by occlusion [7] or other factors. Such low-quality silhouettes can break the continuity of gait sequences and bring in a negative impact on recognition performance, thus we believe that identifying them is important to gait recognition in the wild.

Some prior arts propose absolute-single quality assessment methods to evaluate single silhouette quality [8], performing well on in-the-lab datasets. They calculate one value for each silhouette independently as a quality assessment result. In the wild scenarios, however, the quality of a single silhouette should be determined by the variation state of the entire sequence. For instance, in Fig. 1(b), the first four silhouettes should be given lower quality ratings since the leg is missing. On the other hand, in Fig. 1(c), if severe occlusion like half-body missing presents in the last several silhouettes, the quality ratings of the first several silhouettes should be increased. In other words, we argue that the quality rating should be carried out in a relative manner.

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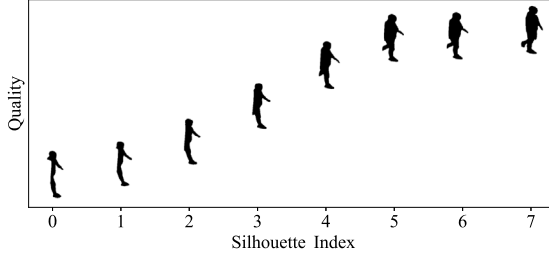
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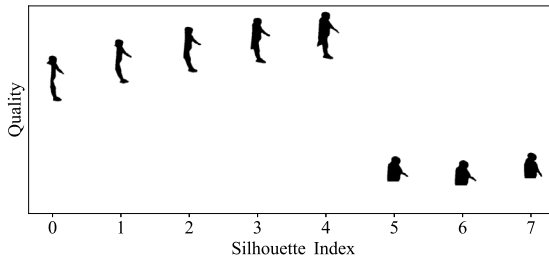
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(a) Gait silhouette sequence.



(b) Quality of the first 8 silhouettes.



(c) Quality of the first 8 silhouettes with part missing.

Fig. 1. Illustration of a silhouette sequence on Gait3D. Silhouettes in (b) and (c) are the first eight silhouettes in (a). Different from (b), the last three silhouettes in (c) miss their lower body.

Inspired by the aforementioned observations, we design a novel gait silhouette quality assessment method, named Average Feature Similarity (AFS). AFS calculates the similarity between a single silhouette and the rest within a sequence to conduct the quality assessment. We believe that gait silhouettes exhibit small differences within a gait cycle if the qualities of these silhouettes are great since pedestrian limb structure and body height remain unchanged. When a silhouette suffers a degradation in quality, the similarity of this silhouette to rest frames changes significantly. Thus, our AFS method assesses the quality by inter-frame/silhouettes similarities. The quality affects how much attention the network should pay to, and improves the performance of gait recognition in the wild consequently.

Based on AFS, we propose a new framework for gait recognition in the wild, named Relative-Sequence Quality Assessment Network (RSANet). After a backbone network for spatial feature extraction, an Average Feature Similarity Module (AFSM) is applied to obtain quality assessment values of silhouettes (AFS sequence) as weights. Then, quality-weighted features are sent into the following 3D blocks for spatio-temporal feature extraction. In 3D blocks, the information on adjacent silhouettes is merged into new feature maps, which enlarges the temporal receptive field. However, the AFS sequence keeps the original temporal receptive field, differing from corresponding feature maps. If the original AFS sequences weight the results of the above 3D blocks, the issue of inconsistent receptive fields arises.

One possible solution is recalculating and updating the AFS sequence after each 3D block. However, the convolution operation maintains the size of feature maps unchanged by zero padding, and this operation changes the receptive field of the edge area. If we apply temporal convolution with zero padding for a silhouette sequence, the temporal receptive fields of silhouettes in the middle of the sequence are larger than the frontmost and rearmost ones. In this work, if we

directly compute the AFS in 3D blocks, it may lead to inconsistent matching between middle silhouettes and two ends. To address this issue, we present a Receptive Field Sustaining Block (RFSB) as a submodule of the AFSM. RFSB contains adaptive convolutions on the AFS sequence, allowing each value in AFS to obtain information from its adjacent frames to increase the temporal receptive field. We name the AFS sequence calculated by RFSB as the 3D AFS sequence. Meanwhile, the computational cost generated by convolution operations is much lower than recalculating the AFS sequence. With the help of RFSB and 3D AFS sequence, RSANet can evaluate a Relative-Sequence quality in a 3D backbone network without inconsistent temporal receptive problems.

Furthermore, due to the large number of parameters and limited temporal receptive field of 3D blocks, we design the Multi-Temporal-Receptive-Field Residual Block (MTB) inspired by [9–12]. MTB consists of two 2D convolution operations and a Temporal Short-Term Dense Concatenate module (Temporal-STDC) performing convolution operations along the temporal dimension. Compared with the P3D residual block [9,10], our MTB has less parameters but a larger temporal receptive field, which reduces the number of parameters to the level of a conventional 2D block [9].

Our RSANet framework achieves 75.0% Rank-1 accuracy on the Gait3D dataset [5] and 79.2% Rank-1 accuracy on the GREW dataset [6] with only 9.89 Megabytes parameters (compared to 27.62 Megabytes parameters on previous methods [11]), showing a better accuracy-cost trade-off than the state-of-the-art methods [11,13–15].

The main contributions are summarized as follows:

- (1) We design a Relative-Sequence Quality Assessment Network (RSANet) to estimate the silhouette quality from the sequence in a relative manner for gait recognition in the wild.
- (2) We propose the Average Feature Similarity Module (AFSM) to calculate the AFS sequence of the silhouette. In addition, we present the Receptive Field Sustaining Block (RFSB) to calculate the 3D AFS sequence in AFSM.
- (3) We propose the Multi-Temporal-Receptive-Field Residual Blocks (MTB) to extend temporal receptive fields with fewer parameters compared to conventional P3D residual blocks.
- (4) Extensive experiments demonstrate that our approach obtains the state-of-the-art performance on in-the-wild datasets.

2. Related work

2.1. Gait recognition

Model-based Methods. Many methods [16–20] consider the physical structure of the human body and utilize pose information as input. For instance, PoseGait [16] uses the pose information to extract temporal-spatial features and employs CNN to extract high-level temporal-spatial features. GaitGraph [17] employs a pose estimator to extract pose features and adopts graph convolutional neural networks [21] for gait recognition. GPGait [19] uses a unified pose representation as input. Then a Part-Aware Graph Convolutional Network is proposed to enable efficient graph partition and local-global spatial feature extraction. Although these methods are robust in some scenarios, such as clothes changed, pose information is not easy to calculate. Therefore, model-based methods may be difficult to apply in a new scenario.

Appearance-based Methods. This category of methods is employed to learn the spatio-temporal variations of gait without explicit body structure. Most methods [11,13,15,22,23] use silhouette sequences as inputs to extract the silhouette feature. Then, the recognition is implemented by comparing sequences. Some methods [24] also use semantic human parsing as inputs, obtaining satisfactory performance.

Previous works [25,26] focus on in-the-lab scenarios and have achieved successful performance. However, with the recent development of in-the-wild datasets, *i.e.*, Gait3D [5] and GREW [6], these methods are facing new challenges. To alleviate these problems, many methods are proposed. GaitGCI [23] uses Counterfactual Intervention Learning to eliminate the impacts of confounders, focusing on the discriminative and interpretable regions effectively. DyGait [13] focuses on the extraction of dynamic features and proposes a Dynamic Augmentation Module to dynamics part features automatically. OpenGait [15] proposes a simple yet effective framework, GaitBase. Based on GaitBase, DeepGaitV2 [11] obtains higher accuracy by increasing the number of stacked blocks. However, it also augments the computational resource requirements. Compared to pose information, appearance information such as silhouette is easy to obtain in a new scenario, having broader adaptability. But every coin has two sides. Appearance information is susceptible to the influence of pedestrian clothes, exhibiting low robustness in certain specific scenarios such as clothes-changing.

Our work pertains to appearance-based methods, utilizing the relative quality of gait to weight silhouette and MTB to extract spatio-temporal information effectively.

Occluded Gait Recognition. Occlusion is one of the key reasons leading to quality degradation of silhouettes. Existing methods can be divided into two categories: *i.e.* Reconstruction-free and Reconstruction-based approaches respectively. The former [27] directly uses machine learning techniques to extract occlusion-robust features from gait sequences. Due to not making use of prior knowledge like holistic body bounding box, such methods cannot tackle severe occlusions.

Reconstruction-based methods [28] try to restore silhouettes or gait features from occluded samples. These methods perform well in some scenarios but require the full-body bounding box even under partial occlusions. To address the aforementioned issue, some methods only using the visible part have been proposed. Xu et al. [29] registers the silhouette based on the estimated occlusion ratio. OccSilGait [30] uses SpaAlign to transform each input silhouette to the inter-media result aligned in spatial, taking both scale and position variations into consideration.

From the perspective of occluded gait recognition, the proposed AFSM can be classified as a reconstruction-free method, assessing quality through inter-frame/silhouette similarities. Unlike conventional reconstruction-free methods, AFSM allocates less attention to the affected parts as the occlusion area increases.

2.2. Biometric image quality assessment

The biometric image quality assessment has witnessed substantial advancements, especially face image quality assessment. For instance, CR-FIQA [31] estimates the face image quality of a sample by learning to predict its relative classifiability, which is measured based on the allocation of the training sample feature representation in angular space to its class center and the nearest negative class center.

But when it comes to gait recognition tasks, few methods estimate the gait image quality and utilize it to assist model training. SQQ [32] explores this domain, analyzing the gait sequence quality based on one dimensional foreground-sum signal modeling and signal processing technique. However, since SQQ relies heavily on stable walking and ideal segmentation, it is challenging to apply it to in-the-wild recognition, where pedestrians exhibit diverse walking patterns and the segmentation performance is poor. GQAN [8] employs FQBlock for assessing single silhouette and PQBlock for predicting scores for each part within the entire silhouette sequence, obtaining promising results on CASIA-B [3] and OU-MVLP [4] datasets. Differently, our method assesses the quality by inter-frames/silhouettes similarities, exploring the impact of relative-sequence quality assessment on in-the-wild datasets [5,6].

3. Method

3.1. Pipeline

To tackle the problem of gait recognition in the wild, we present RSANet, which is composed of the Spatial Feature Extracted Module (SEM), the Spatial-Temporal Feature Extracted Module (STEM), and the Average Feature Similarity Module (AFSM). The pipeline of RSANet is shown in Fig. 2. The silhouette sequences are first fed into SEM to get the spatial features. Then we use these features as input to get the AFS sequence and the 3D AFS sequence in AFSM. We weight the spatial features by using the AFS sequence and feed them into STEM. The output of each block is weighted by the 3D AFS sequence in STEM. Finally, a combined loss function consisting of triplet loss [36] and cross-entropy loss is used to supervise the learning process. In the training stage, all modules work for model training. In the test stage, the loss function is removed. Only the output of the FC layer [34] is used for the evaluation.

3.2. Spatial feature extracted module

The trainable parameters in the Spatial Feature Extracted Module (SEM) contain one Conv2D block (C) and two spatial information extractors 2D Block-1 ($\mathcal{B}_1$) and 2D Block-2 ($\mathcal{B}_2$). For C , it consists of a simple 2D convolution operation with a kernel size of 3×3 . For $\mathcal{B}_1$ and $\mathcal{B}_2$, both consist of 2D residual blocks [9]. Specifically, $\mathcal{B}_1$ contains one block and $\mathcal{B}_2$ contains four blocks. Using $x \in \mathbb{R}^{1 \times T \times H \times W}$ as model input, SEM extracts spatial feature $x_s \in \mathbb{R}^{2C \times T \times \frac{H}{2} \times \frac{W}{2}}$, where C represent channels, T represents the number of silhouettes, H and W represent the height and width respectively. The process can be formulated as

$$x_s = \mathcal{B}_2(\mathcal{B}_1(C(x))). \quad (1)$$

3.3. Spatial-temporal feature extracted module

To calculate the spatial feature to obtain AFS sequence, we use the 2D residual blocks [9] within the SEM without any temporal convolution. However, as a consequence, the current model exhibits a significant reduction in the number of convolutions along the temporal dimension compared to a pure 3D model, leading to the temporal receptive field reduction. One possible solution is to increase the number of 3D residual or P3D residual blocks [9,10]. Another is to directly perform multiple temporal convolution operations in each block. However, these approaches both lead to a significant increase in parameter number, resulting in excessively complex computational requirements. To alleviate the issue mentioned above, we propose a novel spatial-temporal information extractor named Multi-Temporal-Receptive-Field Residual Blocks (MTB) inspired by [9–12], whose architecture is shown in Fig. 2b.

MTB consists of two 2D convolution operations in spatial dimension and a Temporal Short-Term Dense Concatenate module (Temporal-STDC). STDC is originally used to expand the spatial receptive field in 2D semantic segmentation [12]. However, in gait recognition, the issue of receptive field reduction lies in the temporal dimension rather than the spatial dimension (or in spatial size [64, 44]). Therefore, we do not employ STDC in the spatial dimension but rather adopt this concept and implement it in the temporal dimension. Using $x_{in} \in \mathbb{R}^{C \times T \times H \times W}$ as input and $x_t \in \mathbb{R}^{C \times T \times H \times W}$ as output, the process of Temporal-STDC with $n - 1$ ($n \geq 2$) temporal convolution operations is formulated as

$$x_1 = \mathcal{R}(\text{conv1D}_1^t(x_{in})), \quad (2)$$

$$x_i = \mathcal{R}(\text{conv1D}_i^t(x_{i-1})), \quad (3)$$

$$x_t = \text{Cat}(x_1, x_2, \dots, x_n), \quad (4)$$

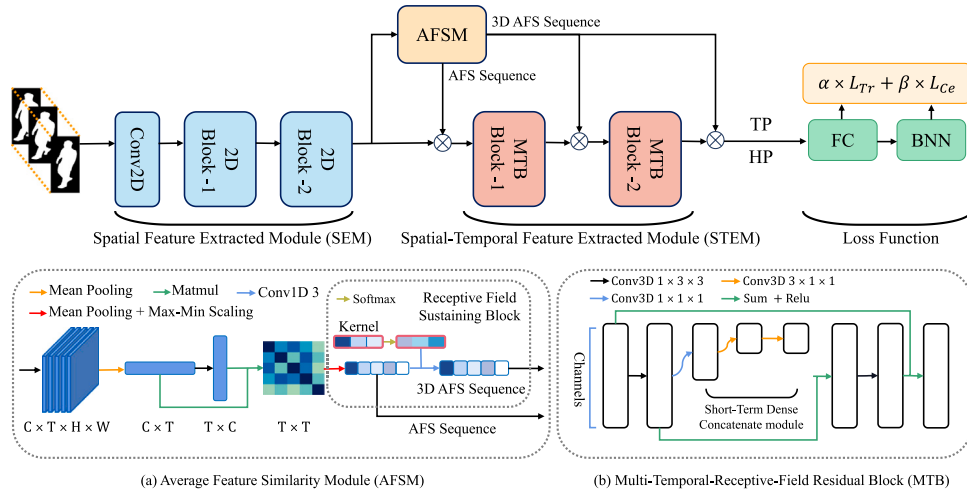


Fig. 2. Pipeline of our RSANet. In the Spatial Feature Extracted Module (SEM), we use the 2D residual block [9] to extract the spatial feature. In the Spatial-Temporal Feature Extracted Module (STEM), Multi-Temporal-Receptive-Field Residual Blocks (MTB) are proposed to extract the spatial-temporal feature. We use the Average Feature Similarity Module (AFSM) to calculate the AFS sequence and the 3D AFS sequence. TP denotes the Temporal Max Pooling operation, HP is the Horizontal Pooling operation [33], FC is the separate fully-connected layers [34], and BNN is BNNeck [35]. The final loss function is composed of a triplet loss [36] and a cross-entropy loss. $\otimes$ is Hadamard Product. Before that, we use a dimension expansion method to ensure the same size of the two inputs.

where $\mathcal{R}$ represents performing BatchNorm3D on x and then activating x with the ReLU activation function, conv1D_1^t is 1D temporal convolution operation with kernel size of 1, conv1D_i^t is 1D temporal convolution with kernel size of k . The number of output channels of conv1D_i^t is $C/2^i$ except for conv1D_n^t , whose number is $C/2^{n-1}$. So the number of output channels of x_i is C .

The size of the temporal-receptive field t obtained through the Temporal-STDC can be calculated by

$$t = r + (n - 1) * (k - 1), \quad (5)$$

where r is the current size of the temporal-receptive field, k is the kernel size in temporal dimension, n is same as in Eq. (4). The stride of temporal convolution operation is 1 in RSANet.

By Eqs. (2), (3) and (4), we can calculate the parameters of Temporal-STDC as follows:

$$\begin{aligned} T_{\text{params}} &= C \times 1 \times \frac{C}{2} + \sum_{i=2}^{n-1} \frac{C}{2^{i-1}} \times k \times \frac{C}{2^i} + \\ &\quad \frac{C}{2^{n-1}} \times k \times \frac{C}{2^{n-1}} \\ &= \frac{C^2}{2} + kC^2 \times \sum_{i=2}^{n-1} \frac{1}{2^{2i-1}} + \frac{kC^2}{2^{2n-2}} \\ &= \frac{C^2}{2} + \frac{kC^2}{6} \times (1 + \frac{1}{2^{2n-3}}) < kC^2. \end{aligned} \quad (6)$$

The parameters of Temporal-STDC are less than a simple 1D temporal convolution operation with a kernel size of k . Since MTB replaces the simple temporal convolution operation with Temporal-STDC, it effectively extends temporal receptive fields even with fewer parameters than a P3D residual block.

Using the Temporal-STDC to extract temporal information, the MTB is given as

$$x_{\text{mid}} = \mathcal{R}(\text{conv2D}_1(x_s)), \quad (7)$$

$$x_{\text{out}} = \mathcal{A}(\text{conv2D}_2(\mathcal{A}(S(x_{\text{mid}}) + x_{\text{mid}})) + x_s), \quad (8)$$

where $\mathcal{R}$ is same as in Eq. (2), S is Temporal-STDC operation, conv2D_1 and conv2D_2 are 2D spatial convolution operations with kernel size of $k \times k$, $\mathcal{A}$ is activation function. Specifically, we use the ReLU activation function. Through MTB, we can effectively increase the receptive field in the temporal dimension while maintaining the parameters within a controllable range, as the parameters of Temporal-STDC are less than a simple 1D temporal convolution operation.

3.4. Average feature similarity module

In this section, we propose an Average Feature Similarity Module (AFSM). We evaluate the silhouette quality by using the Average Feature Similarity (AFS) between a particular silhouette and all other silhouettes. Additionally, we empirically illustrate the significant relationship between the silhouette quality and its AFS.

Average Feature Similarity. In this part, we use the Average Feature Similarity (AFS) to evaluate the silhouette quality. As shown in Fig. 2a, given $x_s \in \mathbb{R}^{2C \times T \times \frac{H}{2} \times \frac{W}{2}}$, an average pooling operation is first applied on the spatial dimension to obtain each silhouette's temporal feature $t \in \mathbb{R}^{2C \times T}$,

$$t = \mathcal{P}_s(x_s), \quad (9)$$

where $\mathcal{P}_s$ is spatial average pooling operation. Then the Silhouette Similarity Matrix ($M \in \mathbb{R}^{T \times T}$) is computed by

$$M = \text{sim}(t^T t), \quad (10)$$

where Tr is transpose and $\text{sim}(\cdot, \cdot)$ is a similarity function, and we use the cosine similarity for all experiments. After that, the AFS sequence s ($s \in \mathbb{R}^T$) is given by

$$z = \mathcal{P}_t(M), \quad (11)$$

$$s = \frac{z - \min(z) + \lambda}{\max(z) - \min(z) + \lambda}, \quad (12)$$

where $\mathcal{P}_t$ is temporal average pooling operation, λ is a hyper-parameter to prevent the occurrence of s being zero.

After obtaining AFSs, we use it to weight the output of SEM, allowing the model to pay more attention to high-quality silhouettes. However, as mentioned in Section 1, there is an inconsistent temporal receptive field between silhouettes and their AFS in STEM. To alleviate this problem, we present the Receptive Field Sustaining Block (RFSB), whose formula is as follows,

$$\bar{k} = \text{softmax}(k/\tau), \quad (13)$$

$$s_{3d} = \text{conv1D}(s, \bar{k}), \quad (14)$$

where k is a learnable parameter with size of 3, conv1D is 1D convolution with the kernel $\bar{k}$, τ is a hyper-parameter to adjust probabilistic distributions.

Via adjusting the number of conv1D , temporal receptive fields of silhouette and its 3D AFS sequence can be aligned. By RFSB, we obtain

Algorithm 1: Backbone of RSANet

Input: Sample $x \in \mathbb{R}^{C \times T \times H \times W}$, 2D-Block-1 (B_1), 2D-Block-2 (B_2), MTB-Block-1 (B_3), MTB-Block-2 (B_4), AFSM ($\mathcal{A}$), RFSB ($\mathcal{R}$).

Output: Sample Feature $f \in \mathbb{R}^{8C \times T \times \frac{H}{4} \times \frac{W}{4}}$.

- 1 $x_s \leftarrow B_2(B_1(x))$, $x_s \in \mathbb{R}^{2C \times T \times \frac{H}{2} \times \frac{W}{2}}$ # Spatial Feature
Extracted Module
- 2 $s \leftarrow \mathcal{A}(x_s)$, $s \in \mathbb{R}^T$
- 3 $s_{3d}^1 \leftarrow \mathcal{R}(s)$, $s_{3d}^1 \in \mathbb{R}^T$
- 4 $s_{3d}^2 \leftarrow \mathcal{R}(s_{3d}^1)$, $s_{3d}^2 \in \mathbb{R}^T$ # AFS sequence
- 5 $f \leftarrow s_{3d}^2 \otimes B_4(s_{3d}^1 \otimes B_3(s \otimes x_s))$, $f \in \mathbb{R}^{8C \times T \times \frac{H}{4} \times \frac{W}{4}}$ #
Spatial-Temporal Feature Extracted Module
- 6 **return** f .

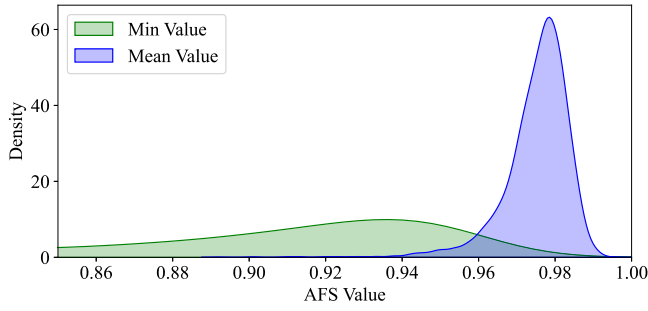


Fig. 3. Kernel density plots for the average and minimum AFS of all silhouettes in each of the silhouette sequences with a length of 50. 6000 sequences in Gait3D [5] are used for test.

s_{3d}^1 and s_{3d}^2 to weight the output of MTB Block-1 (B_3) and MTB Block-2 (B_4) in Fig. 2 respectively. The formulas are as follows:

$$s_{3d}^1 = \text{conv1D}_i(\text{conv1D}_{i-1}(\dots \text{conv1D}_1(s))), \quad (15)$$

$$s_{3d}^2 = \text{conv1D}_j(\text{conv1D}_{j-1}(\dots \text{conv1D}_1(s_{3d}^1))), \quad (16)$$

where i and j are the number of temporal convolution operations in B_3 and B_4 .

By using s , s_{3d}^1 , s_{3d}^2 ($s_{3d}^1, s_{3d}^2 \in \mathbb{R}^T$), x_s (x_s is same as in Eq. (1)) as input, the Spatial-Temporal Feature Extracted Module (STEM) can be formulated as

$$f = s_{3d}^2 \otimes B_4(s_{3d}^1 \otimes B_3(s \otimes x_s)), \quad (17)$$

where $\otimes$ is Hadamard Product, $f \in \mathbb{R}^{8C \times T \times \frac{H}{4} \times \frac{W}{4}}$.

The RSANet backbone procedure is encapsulated in Algorithm 1.

Relationship Between the Silhouette Quality and AFS. To explore the relationship between silhouette quality and AFS, we conduct a statistical analysis by calculating the average and minimum AFS for each of the 6000 silhouette sequences. Then we generate kernel density plots for these values. As illustrated in Fig. 3, the average AFS values are concentrated within the range of 0.96 to 0.98 with a small variance. This indicates that most of the silhouettes in a sequence are quite similar. However, the majority of minimum AFS values are below 0.96 with a large variance. This indicates that there are some abnormal silhouettes that differ significantly from the rest in a silhouette sequence. For such abnormal cases, we assign them a low-quality value. For instance, as shown in Fig. 1(a), the last silhouette only contains a head due to incomplete segmentation, lacking information on the body, legs, and arms compared to other silhouettes, resulting in a low AFS and silhouette quality.

Table 1

Network backbone of the RSANet3D.

Layer name	Output size	Structure
Conv2D	(T, C, 64, 44)	$[3 \times 3, C] \times 1$
2D Block-1	(T, C, 64, 44)	$[3 \times 3, C] \times 1$
2D Block-2	(T, 2C, 32, 22)	$[3 \times 3, 2C] \times 4$
AFSM	(T, 2C, 32, 22)	AFSM
MTB Block-1	(T, 4C, 16, 11)	MTB $\times 4$
MTB Block-2	(T, 8C, 16, 11)	MTB $\times 1$

3.5. Optimization

During the training of RSANet, a combined loss function consisting of triplet loss [$\mathcal{L}_{tr}$] and cross-entropy loss ($\mathcal{L}_{ce}$) is used to supervise the learning process,

$$\mathcal{L} = \alpha \mathcal{L}_{tr} + \beta \mathcal{L}_{ce}, \quad (18)$$

where α and β are hyper-parameters to balance the contributions to the total loss $\mathcal{L}$.

4. Experiments

4.1. Datasets and implementation details

Gait3D [5] is a large gait datasets in the wild. It contains 4000 subjects and 25,309 sequences extracted from 39 cameras. In our experiments, only 2D silhouettes are employed, as shown in Fig. 4c. 3000 subjects are collected as the training set, and the rest 1000 subjects consist of the test set. For the test stage, the probe consists of one sequence of each subject, and the gallery is composed of the rest sequences.

GREW [6] is one of the largest gait datasets in the wild, including Silhouettes shown in Fig. 4d, GEIs, and 2D/3D human poses data types. The raw videos are collected from 882 cameras in large public areas. 7533 video clips with nearly 3500 h of 1080 $\times$ 1920 streams are used. It has 26,345 subjects and 128,671 sequences, divided into two parts with 20,000 and 6000 subjects as training and test sets, respectively. Each subject in the test set has four sequences, two for probe and two for gallery.

BUAA-Duke-Gait [37], which borrows data from DukeMTMC-VideoReID [38], is a new gait dataset for in-the-wild recognition. We present several examples in Fig. 4e. It contains 1404 subjects and 4612 sequences captured from eight different cameras. We use half of the subjects for training and test, respectively. In the test phase, we randomly select one video for each subject in the testing set as a probe while other videos of this subject are in the gallery.

As can be seen from Fig. 4, compared to in-the-lab silhouettes, such as those in CASIA-B [3] and OU-MVLP [4], silhouettes in the wild scenarios exhibit lower quality [5,6,37]. For instance, body-missing in Fig. 4c, blurred-shape in Fig. 4d, and poor-segmentation in Fig. 4e. Using these datasets can provide a comprehensive evaluation.

Implementation Details. All our experiments are implemented using PyTorch [39]. The size of silhouette image for all datasets is 64 $\times$ 44. The hyper-parameter λ in Eq. (12) is set to 0.025 in both Gait3D and Buaa-Duke-Gait, and 0.01 in GREW. τ is set to 0.1 in Eq. (13) and kernel size k is set to 3 in all experiments. In the MTB module, n is set to 3 in Eq. (4). For parameters, we only consider the backbone for all experiments. We design RSANet3D as shown in Table 1. To validate the effectiveness of AFSM on 2D networks, we design another network named RSANet2D based on DeepGaitV2-2D [11] (DGaitV2-2D), adding AFSM without RFSB. We establish RSANet with different parameters: RSANet-T (Tiny), RSANet-B (Base), RSANet-L (Large), and RSANet-H (Huge) to exhibit a cost-accuracy trade-off. They have similar structures except for channels, which are

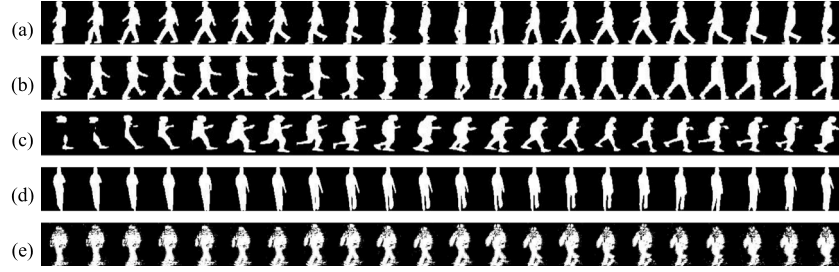


Fig. 4. Exemplar gait sequence of CASIA-B [3] (a), OU-MVLP [4] (b), Gait3D [5] (c), GREW [6] (d), and BUAA-Duke-Gait [37] (e).

(16, 32, 64, 128), (32, 64, 128, 256), (64, 128, 256, 512), and (128, 256, 512, 1024) in every block, respectively. RSANet-Tiny, Base, and Large are trained on a server equipped with 48 Intel(R) Xeon(R) CPU E5-2650 v4 2.20 GHz CPUs and eight NVIDIA GeForce RTX 2080 Ti GPUs. RSANet-Huge is trained with four Intel(R) Xeon(R) CPU E5-2620 v4 2.10 GHz CPUs and two NVIDIA A800 80 GB PCIe GPUs. All inferences are conducted with one Intel(R) Xeon(R) CPU E5-2650 v4 2.20 GHz CPU and NVIDIA GeForce RTX 2080 Ti GPU.

During training, the input size of silhouettes is 64×44 , and silhouettes are ordered in a sequence with a length of 30 on all datasets. The optimizer is the Stochastic Gradient Descent (SGD). We adjust the learning rate, batch size, and the number of iterations to fit different dataset scales. (1) On Gait3D, we train the model for 120k iterations with a batch size of $32 \times 4 \times 30$ (32 pedestrians, each containing four sequences of 30 silhouettes). The learning rate starts at 0.1 and is subsequently decreased by a factor of 0.1 at iterations (40k, 80k, 100k). (2) On GREW, the model is trained for 180k iterations with a batch size of $32 \times 4 \times 30$. The learning rate starts at 0.05 and is subsequently decreased by a factor of 0.2 at iterations (60k, 120k, 150k). (3) On BUAA-Duke-Gait, we conduct training for 120k iterations with a batch size of $32 \times 4 \times 30$. The learning rate, set at 0.1, is reduced by a factor of 0.1 at iterations (40k, 80k, 100k).

During testing, all silhouettes in a sequence are fed into the model. Specifically, we train the models in GaitBase [15] and DGaitV2 [11] by ourselves. As shown in Table 2, the results on the left side of the parentheses represent our results, while the results inside the parentheses originate from the original paper [11,15].

4.2. Performance comparison

Gait3D. We compare our proposed RSANet with other methods on the Gait3D dataset. The results are reported in Table 2. RSANet3D-B improves 1.5% Rank-1 accuracy than DGaitV2-3D-B using less than half the parameters. Even RSANet3D-T improves 2.2% and 15.5% Rank-1 accuracy than HSTL and DANet using only 0.77 MegaBytes parameters. RSANet3D-L and RSANet3D-H both outperform DGaitV2-P3D with the same channels with quality-assessment module AFSM and effective spatial-temporal feature extracted module MTB.

We observe that earlier methods, *i.e.*, GaitGL [25], GaitPart [26] perform poorly on Gait3D. The possible reason is that they exhibit a small receptive field. According to our calculations, the temporal-spatial receptive fields of GaitGL and GaitPart are (19, 27, 27) and (8, 37, 37) respectively. For an input size of (30, 64, 44), they may be considered inadequate. However, this problem has been addressed in RSANet3D. MTB has the same spatial receptive field (119, 119) as the P3D block does. Differently, MTB leverages Temporal-STDC to expand the temporal receptive field without parameter increase, which can capture more information from a silhouette sequence.

GREW. The comparison on the one of largest in-the-wild dataset GREW is conducted in Table 2. We observe that RSANet3D exhibits a higher increase in accuracy on the GREW dataset compared to the Gait3D dataset under the same parameter increase. The possible reason is that GREW has 26,345 subjects and 128,671 sequences. They are far

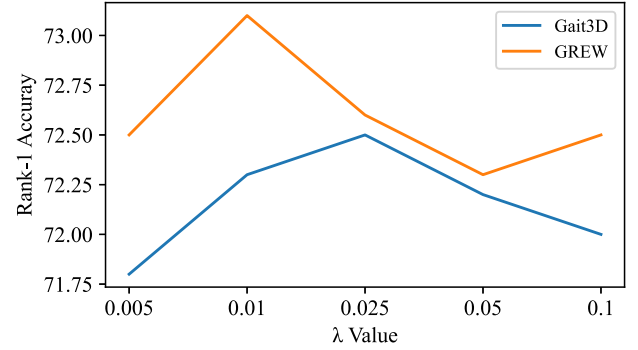


Fig. 5. Ablation study of hyper-parameter λ in both Gait3D [5] and GREW [6].

more than Gait3D which contains 4000 subjects and 25,309 sequences. The larger the model, the more it can learn from GREW. Nevertheless, RSANet3D-B still outperforms the latest methods [13–15] using 3.00 MegaBytes parameters. Finally, the RSANet-3D-H achieves a state-of-the-art result, 81.8%, even outperforming DGaitV2-3D-H with less than half parameters.

BUAA-Duke-Gait. We also evaluate RSANet on BUAA-Duke-Gait dataset in Table 3. Compared to Gait3D and GREW, BUAA-Duke-Gait is relatively small-scale since it contains only 4612 sequences. Even though, compared with other methods, RSANet2D-B and RSANet3D-B exceed DGaitV2-2D-B and DGaitV2-P3D-B1 by 1.1% and 0.8% in Rank-1 accuracy, respectively. RSANet3D-L obtains superior performance, obtaining a Rank-1 accuracy of 77.6%. The results demonstrate that RSANet is robust to overfitting.

Time consuming. We evaluate the runtime of RSANet of different sizes using a sequence of 30 silhouettes (64×44) as input. The results are shown in Table 4. We can observe: (1) Due to time consumed by data processing and feed-forward operations, RSANet-2D-T only decreases by 0.6 ms to 2D-B. It is the same when applies to 3D-T. (2) 3D models consume more time than 2D models due to the temporal convolution operations. (3) Compared with huge models, large models achieve similar accuracy in Table 2, but with only one-third of the runtime and a quarter of parameters. Finally, we consider large models as the main model for a reasonable cost-accuracy trade-off.

4.3. Ablation study

The value of hyper-parameter λ . To find appropriate λ in Eq. (12) on different datasets, we conduct the ablation experiments illustrated in Fig. 5. The result indicates that the optimal λ value varies across datasets. For high-quality datasets, where the number of low-quality silhouettes is limited, a small λ enhances the model's ability to learn the differentiation between high-quality and low-quality silhouettes. Conversely, a big λ is necessary to prevent the model from focusing excessively on high-quality silhouettes, which may affect the extraction of temporal features from the sequences. Finally, we chose a λ of 0.025

Table 2

Rank-1 accuracy, Rank-5 accuracy, Rank-10 accuracy, mean Average Precision (%), and Parameters (M) on Gait3D [5] and GREW [6] datasets.

Method	Gait3D			GREW			Params (M)
	Rank-1	Rank-5	mAP	Rank-1	Rank-5	Rank-10	
GaitGL [25]	29.7	48.5	22.3	47.3	63.6	69.3	2.59
GaitPart [26]	28.2	47.6	21.6	44.0	60.7	67.3	1.47
SMPLGait [5]	46.3(42.9 ^c)	64.5(63.9 ^c)	37.2(35.2 ^c)	–	–	–	–
DANet [2]	48.0	69.7	–	–	–	–	–
DyGait [13]	66.3	80.8	56.4	71.4	83.2	86.8	–
HSTL [14]	61.3	76.3	55.5	62.7	76.6	81.3	4.05
GaitBase [15]	64.3 (64.6 ^b)	79.6	55.5	59.1 (60.1 ^b)	74.5	78.9	4.90
HybridGait [40]	53.3 (42.9 ^c)	72.0	43.3 (35.2 ^c)	–	–	–	23.21
AttenGait [41]	–	–	–	70.7 (50.2 ^c)	82.9 (67.3 ^c)	86.9 (73.4 ^c)	–
CLTD [42]	69.7	85.2	–	78.0	87.7	–	–
CLASH [43]	52.4 (45.3 ^c)	69.2	40.2	67.0 (54.7 ^c)	78.9	83.0	–
DGaitV2-2D-B [11]	64.5 (62.9 ^a)	81.7	56.5	62.3 (62.2 ^a)	76.4	81.5	2.35
DGaitV2-2D-L [11]	67.8 (68.2 ^a)	83.9 (84.6 ^a)	59.7 (60.4 ^a)	69.7 (68.6 ^a)	82.4 (82.0 ^a)	86.7 (86.1 ^a)	9.33
DGaitV2-2D-H [11]	69.5 ^a	–	–	71.8 ^a	–	–	37.30
DGaitV2-P3D-B [11]	70.8 (67.9 ^a)	85.7	62.9	72.6 (72.8 ^a)	84.5	87.9	2.79
DGaitV2-P3D-L [11]	74.2 (74.4 ^a)	86.9 (88.0^a)	67.1 (65.8 ^a)	78.3 (77.7 ^a)	88.5 (87.9 ^a)	91.4 (90.6 ^a)	11.12
DGaitV2-P3D-H [11]	75.0 ^a	–	–	81.0 ^a	–	–	44.43
DGaitV2-3D-B [11]	71.0 (69.4 ^a)	85.0	62.3	73.1 (73.1 ^a)	84.9	88.4	6.92
DGaitV2-3D-L [11]	74.1 (72.8 ^a)	87.0 (86.2 ^a)	66.5 (63.9 ^a)	79.0 (79.4 ^a)	88.9 (88.9 ^a)	91.6 (91.4 ^a)	27.62
DGaitV2-3D-H [11]	75.7 (75.8^a)	87.6	67.6	80.6 (81.6 ^a)	89.7	92.2	110.44
RSANet2D-T (ours)	58.2	76.3	48.9	56.9	72.0	77.3	0.59
RSANet2D-B (ours)	66.3	83.1	57.6	64.0	78.2	82.7	2.35
RSANet2D-L (ours)	68.6	83.1	60.4	70.0	82.8	86.7	9.33
RSANet2D-H (ours)	70.5	84.6	62.1	72.8	84.8	88.4	37.30
RSANet3D-T (ours)	63.5	81.5	54.1	62.3	76.2	81.2	0.64
RSANet3D-B (ours)	72.5	86.1	63.8	73.1	84.9	88.2	2.48
RSANet3D-L (ours)	75.0	87.5	66.2	79.2	89.0	91.8	9.89
RSANet3D-H (ours)	75.2	86.5	66.4	81.8	90.8	93.1	39.51

^a Reported in the original paper [11].^b Reported in the original paper [15].^c Using silhouette sequence as input.**Table 3**

Rank-1, Rank-5, and Rank-10 accuracy on BUAA-Duke-Gait [37] dataset.

Method	Rank-1	Rank-5	Rank-10
GaitSet [34]	44.4	56.1	63.3
GaitPart [26]	24.3	32.1	37.4
RealGait [37]	57.9	71.6	75.6
GaitBase [15]	62.9	72.5	77.6
DGaitV2-2D-B [11]	71.0	80.0	84.4
DGaitV2-2D-L [11]	72.0	81.0	84.2
DGaitV2-P3D-B [11]	75.9	84.1	87.9
DGaitV2-P3D-L [11]	76.7	85.8	88.7
RSANet2D-B	72.1	80.1	83.8
RSANet2D-L	73.6	81.7	85.6
RSANet3D-B	76.7	83.7	87.3
RSANet3D-L	77.6	85.5	89.0

Table 4

The time consuming of RSANet with millisecond (ms). T, B, L, and H represent tiny, base, large, and huge, respectively.

Method	T	B	L	H
RSANet-2D	3.3	3.9	6.8	17.8
RSANet-3D	7.0	8.6	13.1	39.7

Table 5Ablation study of hyper-parameters (α , β) in Eq. (18) on Gait3D [5] with Rank-1 (%).

Method	(0.0, 1.0)	(1.0, 0.0)	(0.5, 1.0)	(1.0, 0.5)	(1.0, 1.0)
RSANet2D-B	63.4	51.5	63.9	64.2	66.3
RSANet3D-B	68.8	56.0	69.9	70.9	72.5

for Gait3D [5] and 0.01 for GREW [6]. The results also indicate that Gait3D has lower quality than GREW.

Table 6

Ablation study of AFSM on Gait3D [5] and GREW [6] with Rank-1 accuracy (%) and Params (M).

Method	Gait3D	GREW	Params (M)
RSANet2D-T w/o AFSM	57.0	55.3	0.59
RSANet2D-T w/ AFSM	58.2	56.9	0.59
RSANet2D-B w/o AFSM	64.5	62.3	2.35
RSANet2D-B w/ AFSM	66.3	64.0	2.35
RSANet3D-T w/o AFSM	62.6	61.4	0.64
RSANet3D-T w/ AFSM	63.5	62.3	0.64
RSANet3D-B w/o AFSM	71.2	71.0	2.48
RSANet3D-B w/ AFSM	72.5	73.1	2.48

Effectiveness of triplet and cross-entropy loss. As shown in Table 5, when α and β are equal, we can balance the total loss well. Triplet loss (Tri) can reduce the distance between positive samples and increase the distance between negative samples, thus reducing the intra-class distance. Since the cross entropy loss (CE) requires a classifier (BNNeck in Fig. 2), it indirectly constructs the class-center of each category, making the classification model stable. Therefore, both Tri and CE are indispensable.

Effectiveness of AFSM.

(1) To verify the effectiveness of AFSM directly, we conduct ablation experiments on RSANet2D and RSANet3D. As shown in Table 6, AFSM improves both RSANet2D and RSANet3D. The results indicate that AFSM provides a way to facilitate information interaction in the entire silhouette sequence, enabling the model to identify low-quality silhouettes and focus more attention on high-quality ones.

(2) We compare the effectiveness of silhouette quality evaluation between AFSM and GQAN [8] in Table 7. We add both FQBlock [8] and PQBlock [8] in GQAN [8] to DGaitV2-2D-B [11] and RSANet3D-B to compare GQAN [8] with AFSM. Different from evaluating a single silhouette quality using FQBlock, GQAN [8] uses PQBlock to predict

Table 7

Comparison between AFSM and GQAN [8] with Rank-1 accuracy (%) and Params (M) using DGaitV2-2D-B [11] and RSANet3D-B as backbone on Gait3D dataset.

Method	FQBlock	PQBlock	AFSM	Rank-1
DGaitV2-2D-B	-	-	-	64.5
	✓	-	-	63.6
	-	✓	-	64.9
	✓	✓	-	63.9
	-	-	✓	66.3
RSANet3D-B	-	-	-	71.2
	✓	-	-	70.1
	-	✓	-	71.0
	✓	✓	-	70.7
	-	-	✓	72.5

Table 8

Ablation study of the weighting places on Gait3D dataset, where Beginning, Middle, and End indicate the places of Hadamard Product in Fig. 2.

Method	Beginning	Middle	End	Rank-1
RSANet2D-B	✓	-	-	64.5
	✓	✓	-	65.4
	✓	-	✓	65.3
	-	✓	✓	65.6
	✓	✓	✓	66.3

Table 9

Ablation on different Quality Values (QV) with Rank-1 accuracy (%) on Gait3D dataset.

Method	QV<0.97	QV≥0.97	All
RSANet3D w/o AFSM	67.9	72.7	71.2
RSANet3D w/ AFSM	71.6	73.0	72.5

a score for each part to compute the weighted distance between the probe and gallery. For DGaitV2-2D-B, PQBlock enhances Rank-1 accuracy by 0.4%. However, the integration of both FQBlock and PQBlock results in 0.6% and 0.5% Rank-1 accuracy decrease to DGaitV2-2D-B and RSANet3D-B, respectively. The performance dropping out may be caused by these blocks, which are tailored specifically for in-the-lab datasets and 2D models, making them unsuitable for direct application to in-the-wild datasets and 3D models. Finally, AFSM outperforms GQAN [8] by 2.4% on DGaitV2-2D-B and by 1.8% on RSANet3D-B, demonstrating its effectiveness.

(3) We conduct ablation experiments on the weighted positions of AFSM in Table 8. When using AFS to weight features at all places, it obtains the best result. The reason may lie in that the network has the property of forgetting. If we only use AFS to weight the features in one place, the network will forget its function after several layers. Therefore, we use AFS to weight the features at several places to remind the network.

(4) To verify the effectiveness of AFSM on low-quality data, as shown in Fig. 3, we categorize the test data into two groups based on the average silhouette quality value (QV) of the sequences: low quality (QV<0.97) and high quality (QV≥0.97) in Table 9. By contrast, although RSANet3D-B with AFSM only improves by 1.5% compared to the same model without it on all test data, AFSM can significantly improve the accuracy on low-quality data, surpassing RSANet3D-B without it by 3.7%. This result proves that AFSM can effectively detect low-quality silhouettes, assisting the model in obtaining good performance on low-quality datasets.

(5) To verify the consistency across different datasets, we conduct ablation experiments with AFSM on Gait3D [5] and GREW [6]. Specifically, we evaluate AFSM trained from one dataset on another dataset and vice versa. As shown in Table 10, AFSM improves the performance of RSANet2D-B and RSANet3D-B in both Gait3D → GREW and GREW →

Table 10

Ablation on consistency of AFSM with Rank-1 accuracy (%) across Gait3D and GREW dataset.

Method	Gait3D ↓ GREW	GREW ↓ Gait3D
RSANet2D-B w/o AFSM	31.6	27.4
RSANet2D-B w/ AFSM	32.8	29.1
RSANet3D-B w/o AFSM	35.2	31.5
RSANet3D-B w/ AFSM	35.4	31.8

Table 11

Ablation study of normalization with Rank-1 accuracy (%) on Gait3D dataset.

Method	AFSM	FQBlock	Norm	Rank-1
DGaitV2-2D-B	-	-	-	64.5
	-	✓	-	63.6
	-	✓	✓	64.2
	✓	-	-	65.3
	✓	-	✓	66.3
RSANet3D-B	-	-	-	71.2
	-	✓	-	70.1
	-	✓	✓	70.8
	✓	-	-	71.6
	✓	-	✓	72.5

Table 12

Ablation study of AFSM, MTB, and RFSB on Gait3D dataset with Rank-1 accuracy (%) and Params (M).

Method	AFSM	MTB	RFSB	Rank-1	Params (M)
RSANet3D-B	-	-	-	70.7	2.73
	✓	-	-	71.3	2.73
	✓	✓	-	71.8	2.48
	✓	✓	✓	72.5	2.48

Gait3D across-dataset tasks. The results demonstrate AFSM can ensure consistency across different datasets.

Effectiveness of sequence quality and the relative manner. RSANet evaluates the quality from sequence by Eqs. (9), (10), and (11); and uses the quality in a relative manner via normalization (Eq. (12)). To prove their effectiveness, we conducted ablation experiments on normalization as shown in Table 11. Without normalization, AFSM enhances the Rank-1 accuracy of DGaitV2-2D-B by 0.8% and RSANet3D-B base model by 0.4%. This demonstrates that evaluating quality from sequence is effective. Furthermore, normalization improves the performance of AFSM and FQBlock in both models, proving the significance of a relative manner. Normalization constrains the impact of quality on the model within an appropriate range, enhancing the model's stability.

Effectiveness of MTB. In Table 12, to demonstrate the effectiveness of MTB, we replace MTB with a regular P3D Block [9–11], and the result is shown in the second row. By comparing results in the third and fourth rows, MTB improves 0.5% Rank-1 accuracy than P3D, using 0.26 MegaBytes parameters less than it. By our calculations, the maximum temporal receptive field of RSANet3D-B is 21, while that of RSANet3D-B without MTB is only 11. This observation demonstrates the significance of the temporal receptive field for gait recognition in the wild. RSANet3D obtains a large temporal receptive field via MTB, thus performing well on in-the-wild datasets.

Effectiveness of RFSB. We validate the effectiveness of RFSB in the last row of Table 12. RFSB improves 0.7% Rank-1 accuracy with less than 1e-2 MegaBytes parameters increase, proving the importance of the temporal receptive field between silhouette and its AFS sequence. RFSB performs adaptive convolution on the AFS sequence, allowing each value in the AFS sequence to represent the quality of multiple interconnected silhouettes.

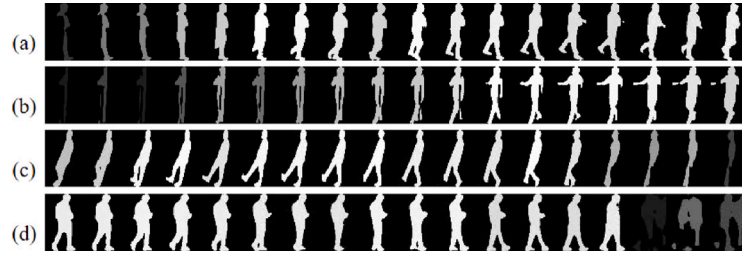
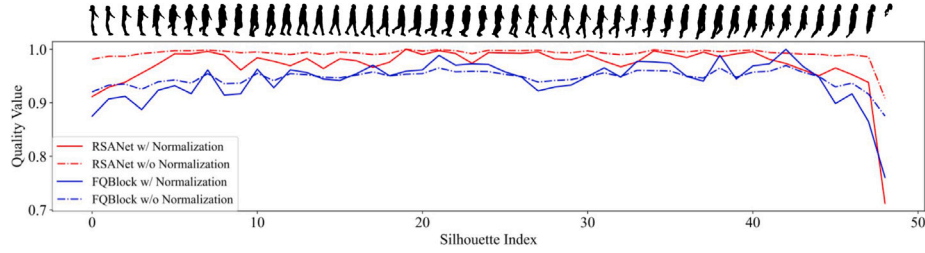
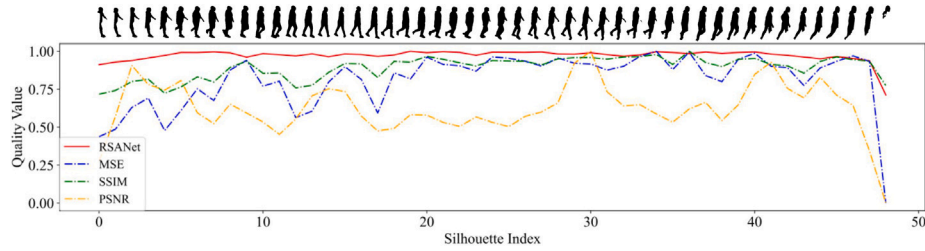


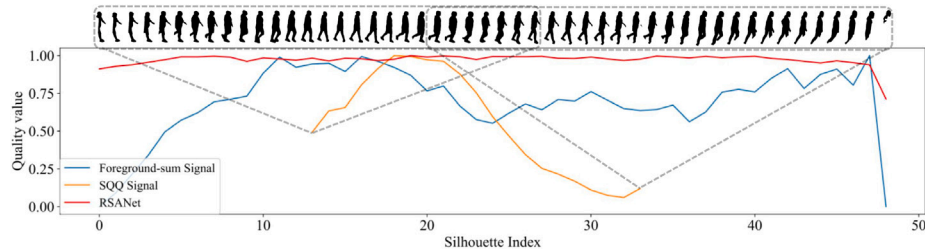
Fig. 6. Silhouette quality evaluated by AFSM. The degree of whiteness is the quality level.



(a) Comparison with FQBlock [8].



(b) Comparison with model-free silhouette quality estimation methods.



(c) Comparison with SQQ [32].

Fig. 7. Comparison our AFSM with conventional silhouette quality estimation methods (a), (c) and FQBlock [8] (b), where the pedestrian is from Gait3D [5] '0294 camid32_videoid3 seq0'.

4.4. Visualization

(1) We conduct the silhouette quality visualization to verify the effectiveness of AFSM on Gait3D [5] as illustrated in Fig. 6. For occlusion (Figs. 6a and 6b), blur (Fig. 6c), and segmentation errors (Fig. 6d), AFSM can identify them and assign them lower quality scores. Consequently, the performance of RSANet can be improved by focusing on high-quality silhouettes.

(2) We compare AFS with other silhouette quality assessment methods by evaluating the silhouette quality in Fig. 1(a). The result is illustrated in Fig. 7.

In Fig. 7(a), we compare AFSM with FQBlock [8] and show the function of relative manner (normalization). AFSM with normalization exhibits greater stability than FQBlock and is sensitive to occluded silhouettes such as the first and last ones. However, AFSM without normalization cannot effectively differentiate between high-quality and slightly low-quality silhouettes, exemplified by the second

and third ones. Therefore, we consider that a relative manner, such as normalization, is essential for sequence quality assessment methods.

In Fig. 7(b), we compare AFSM with conventional model-free methods, including Mean Squared Error (MSE), Structural Similarity Measure (SSIM [44]), and Peak Signal Noise Ratio (PSNR). AFSM assigns higher scores to high-quality silhouettes and lower scores to low-quality silhouettes. Other methods assign lower scores to low-quality silhouettes but perform poorly in identifying high-quality silhouettes.

In Fig. 7(c), we compare AFSM with a conventional sequence silhouette quality quantification method [32] (SQQ) using normalization. Since SQQ computes the difference between harmonics energy and noise energy within a sliding window, the value of each point calculated by SQQ represents the overall quality of the sequence in the subsequent sliding window. Specifically, we select a gait cycle $T = 28$ as the sliding window size in Fig. 7(c). SQQ assigns high scores to sequences of high quality, such as silhouettes from index 5 to index 33, and low scores to sequences of low quality, such as index

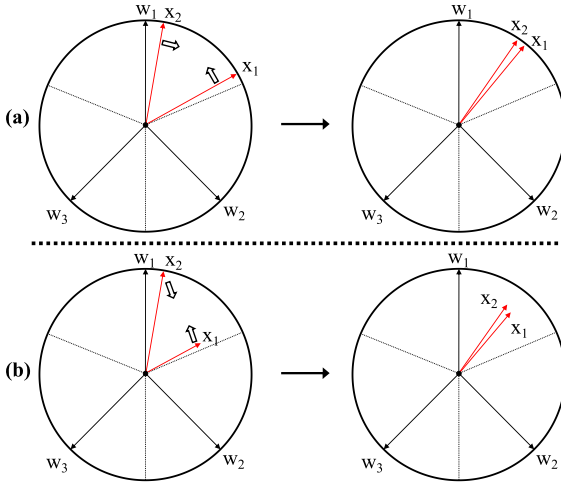


Fig. 8. Feature norm and gradient of two kinds of special low-quality cases, where in (a) all silhouettes are the same and in (b) all silhouettes are significantly different. w_i is class center in BNNeck [35], x_1 is low-quality case feature, x_2 is normal case feature, dashed lines in the circle are class boundaries, $\Rightarrow$ is the gradient of the feature.

19 to index 47. However, it cannot predict the quality of individual silhouettes and is sensitive to foreground-sum signals. For instance, the last silhouette lacks most of the body, resulting in a small foreground-sum. This foreground-sum decline paradoxically enhances the energy of the harmonics, leading to an increase in the values calculated by SQQ. Neither of these issues occurs with AFSM.

4.5. Discussion

(1) Discussion of AFSM on special cases. In low-quality silhouette sequences, two extreme cases may occur in the wild datasets. One is that all silhouettes are the same and another is that all silhouettes are significantly different.

For the first case, AFSM computes a relative-high value for each silhouette, resulting in the feature norm after the temporal pooling phase being similar to a normal case as illustrated in Fig. 8a. For another, AFSM computes a low value for each one, leading the feature norm to be smaller than a normal case as shown in Fig. 8b. Differently, absolute-single quality assessment methods, such as FQBlock [8], assign a low value for each silhouette in both cases. However, both AFSM and FQBlock fail to address two cases above. In Fig. 8, due to the gradients generated by the triplet loss [36], normal case x_2 diverges from the class center and moves closer to the bad case x_1 , leading to an adverse effect on the model's ability to differentiate from the class center.

(2) Discussion on temporal information. As a biological characteristic, gait describes the posture of walking, which is a sequence of movements constituting a locomotion process. It contains both spatial information in terms of morphology and temporal information regarding variations in motion. Therefore, temporal information needs to be emphasized in gait recognition.

We observe that DGaitV2-P3D-B [11] improves 6.3% and 10.3% Rank-1 accuracy than DGaitV2-2D-B on Gait3D [5] and GREW [6] datasets with 0.45 MegaBytes parameters increase in Table 2. This observation implies the importance of temporal information for gait recognition in the wild. However, we also observe that DGaitV2-3D-B [11] improves 6.5% and 10.8% Rank-1 accuracy than DGaitV2-2D-B on the above two datasets with 4.57 MegaBytes parameters increase in Table 2. In comparison to DGaitV2-P3D-B [11], DGaitV2-3D-B [11] exhibits an increase of nearly 10 times in parameters, but the accuracy improvement is limited. This phenomenon indicates that, compared to 2D gait models, the performance improvement of 3D gait models lies not primarily in the increase of parameters, but rather in the ability

to capture temporal gait information. Based on this, we devise the RSANet3D that enables more efficient extraction of temporal information. Besides, RSANet3D can also compute silhouette quality to assist in training. Finally, RSANet3D-B improves 8.0% and 10.8% Rank-1 accuracy than DGaitV2-2D-B on the above two datasets with only 0.13 MegaBytes parameters increase in Table 2, proving its effectiveness.

We believe that using temporal information is essential in subsequent work on gait recognition tasks. Employing a few additional parameters to extract temporal information significantly improves the accuracy, presenting an excellent trade-off between computational cost and accuracy.

5. Conclusions

5.1. Summary

In this paper, we propose a Relative-Sequence Quality Assessment Network (RSANet) for gait recognition. RSANet can evaluate relative silhouette quality by the Average Feature Similarity Module (AFSM). Furthermore, RSANet uses the Multi-Temporal-Receptive-Field Residual Blocks (MTB) to extend temporal receptive fields, improving the performance while using fewer parameters compared with conventional P3D residual blocks. Since our ASFM can be detached from the network and incorporated into a new network, we expect it to aid a more efficient model in achieving superior results. This work can also be applied to widespread surveillance gait recognition systems, assisting in selecting high-quality silhouette sequences. Extensive experiments have been conducted on Gait3D, GREW, and BUAA-Duke-Gait. The results demonstrate that our approach outperforms state-of-the-art methods on in-the-wild datasets.

5.2. Limitation

(1) In theory, AFSM cannot tackle two extremely low-quality cases. The one in which all silhouettes are the same and another where all silhouettes are significantly different. The reason has been analyzed in Section 4.5 (1).

(2) For pure 2D models and 2D-3D models, AFSM can be directly applied as a plug-in. However, AFSM cannot be directly integrated to 3D blocks. This issue restricts the compatibility of AFSM.

5.3. Future work

(1) We consider recognizing and discarding two extremely low-quality cases above in the training stage. For instance, in the context of triplet loss, we can design a threshold, and if the distance between positive samples exceeds it, they are discarded.

(2) Leveraging generation models such as stable-diffusion [45] to complement the low-quality silhouettes is another possible direction. Since AFSM can identify low-quality silhouettes, we can use AFSM as a pre-processing operation rather than a part of models. Specifically, we first use AFSM to identify low-quality silhouettes. Then we feed it to generation models to restore a high-quality one. By replacing the low-quality frames, the performance is expected to be improved. In other words, the robustness of gait model can be improved.

CRedit authorship contribution statement

Guozhen Peng: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yunhong Wang:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis. **Shaoyong Zhang:** Writing – review & editing, Writing – original draft, Supervision, Software, Methodology, Formal analysis. **Rui Li:** Writing – review & editing, Supervision, Software,

Methodology. **Yuwei Zhao:** Writing – review & editing, Methodology, Investigation. **Annan Li:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Software, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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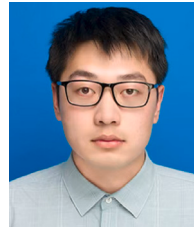
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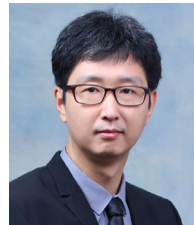
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